Prob 1 - PDE'S

a) $$u = \frac{\sin(2\pi x)}{16\pi^4 - 4\pi^2}, \quad x \in [0,1]$$

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x)$$

$$\partial_x u = \frac{2\pi \cos(2\pi x)}{16\pi^4 - 4\pi^2}$$

$$\partial_x^2 u = \frac{-4\pi^2 \sin(2\pi x)}{16\pi^4 - 4\pi^2}$$

$$\partial_x^3 u = \frac{-8\pi^3 \cos(2\pi x)}{16\pi^4 - 4\pi^2}$$

$$\partial_x^4 u = \frac{16\pi^4 \sin(2\pi x)}{16\pi^4 - 4\pi^2}$$

$$\partial_x^4 u + \partial_x^2 u = \frac{16\pi^4 \sin(2\pi x)}{16\pi^4 - 4\pi^2} + \frac{-4\pi^2 \sin(2\pi x)}{16\pi^4 - 4\pi^2} = \frac{\cancel{16\pi^4 - 4\pi^2}}{\cancel{16\pi^4 - 4\pi^2}} \sin(2\pi x)$$

$u(0) = u(1) = 0 \Rightarrow$ OK!

b) $$u = \frac{\sin(2\pi x)}{16\pi^4 - 4\pi^2} + \alpha \sin(4\pi x), \quad x \in [0,1]$$

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x) + \sin(4\pi x)$$

$$\partial_x u = \frac{2\pi \cos(2\pi x)}{16\pi^4 - 4\pi^2} + 4\pi \alpha \cos(4\pi x)$$

$$\partial_x^2 u = \frac{-4\pi^2 \sin(2\pi x)}{16\pi^4 - 4\pi^2} - 8\pi^2 \sin(4\pi x)$$

$$\partial_x^3 u = \frac{-8\pi^3 \cos(2\pi x)}{16\pi^4 - 4\pi^2} - 16\pi^3 \cos(4\pi x)$$

$$\partial_x^4 u = \frac{16\pi^4 \sin(2\pi x)}{16\pi^4 - 4\pi^2} + 32\pi^4 \sin(4\pi x)$$

$$\partial_x^4 u + \partial_x^2 u = \frac{16\pi^4 \sin(2\pi x)}{16\pi^4 - 4\pi^2} - 8\pi^2 \sin(4\pi x) + \left(\frac{-4\pi^2 \sin(2\pi x)}{16\pi^4 - 4\pi^2} + 32\pi^4 \sin(4\pi x)\right)$$

$$= \sin(2\pi x) + (32\pi^4 - 8\pi^2)\sin(4\pi x) \Rightarrow \underline{\alpha = \frac{1}{32\pi^4 - 8\pi^2}}$$

Prob 2 - Superposition

$$\partial_t - \partial_{xx} u = f(x), \quad u_1 = e^{-4\pi^2 t} \cos(2\pi x) \quad u_2 = e^{-16\pi^2 t} \cos(4\pi x)$$

a) $$\partial_t u_1 = -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)$$

$$\partial_x u_1 = -2\pi e^{-4\pi^2 t} \sin(2\pi x)$$

$$\partial_x^2 u_1 = -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)$$

$$\partial_t u_1 - \partial_{xx} u_1 = -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x) - (-4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)) = 0 = f(x)$$

$$\partial_t u_2 = -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)$$

$\partial_x u_2 = -4\pi e^{-16\pi^2 t} \sin(4\pi x)$

$\partial_{xx} u_2 = -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)$

$\partial_t u_2 - \partial_{xx} u_2 = -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x) - (-16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)) = 0 = f(x) \Rightarrow OK!$

b) Superpos. states if both x and y are a solution to the eq, x+y will also solve the eq.

c) $f(x) = \sin(2\pi x)$

$u_3 = \frac{1}{4\pi^2}(1 - e^{-4\pi^2 t}) \sin(2\pi x) = \sin(2\pi x) - e^{-4\pi^2 t} \sin(2\pi x)$

$\partial_t u_3 = \frac{1}{4\pi^2}(1 + 4\pi^2 e^{-4\pi^2 t}) \sin(2\pi x) = e^{-4\pi^2} \sin(2\pi x)$

$\partial_x u_3 = \frac{1}{2\pi}(1 - e^{-4\pi^2 t}) \cos(2\pi x)$

$\partial_{xx} u_3 = (e^{-4\pi^2 t} - 1) \sin(2\pi x)$

$\partial_t u_3 - \partial_{xx} u_3 = e^{-4\pi^2} \sin(2\pi x) - (e^{-4\pi^2 t} - 1) \sin(2\pi x) = \underline{\sin(2\pi x)} = f(x)$

Since $f(x) \neq 0$, the eq is no longer homogeneous. For the superpos. to be valid this needs to hold.

Hence $u_3 + u_3$ is not a solution

d) $u_3(0, x) = 0 \Rightarrow ok$

$u_3(t, 0) = u_3(t, 1) = 0 \Rightarrow ok$

e) $u_4 = e^{-16\pi^2 t} \sin(4\pi x)$

$\partial_t u_4 = -16 e^{-16\pi^2 t} \sin(4\pi x)$

$\partial_x u_4 = 4\pi e^{-16\pi^2 t} \cos(4\pi x)$

$\partial_{xx} u_4 = -16\pi e^{-16\pi^2 t} \sin(4\pi x)$

$\partial_t u_4 - \partial_{xx} u_4 = -16 e^{-16\pi^2 t} \sin(4\pi x) - (-16 e^{-16\pi^2 t} \sin(4\pi x)) = 0 = f(x) \Rightarrow ok$

$u(0, x) = \sin(4\pi x) = g(x) \Rightarrow ok!$

Prob 3

a) $u_t = u_{xx}$ linear and homogeneous

b) $u_t = (x^2 - x) u_{xx}$ linear and homogeneous

c) $r_r u_{rr} + \frac{1}{r} u_{\theta\theta} = 0$ linear and homogeneous

d) $u_{xx} + u_{yy} + 3 = 0$ linear and inhomogeneous

e) $u_t + u_x + u^n = 0$ linear for $n=1$, else nonlinear and homogeneous

f) $u_{xxxx} + 8 u_{xxyy} + u_{yyyy} = 0$ linear and homogeneous